29/10/26

Signal Processing

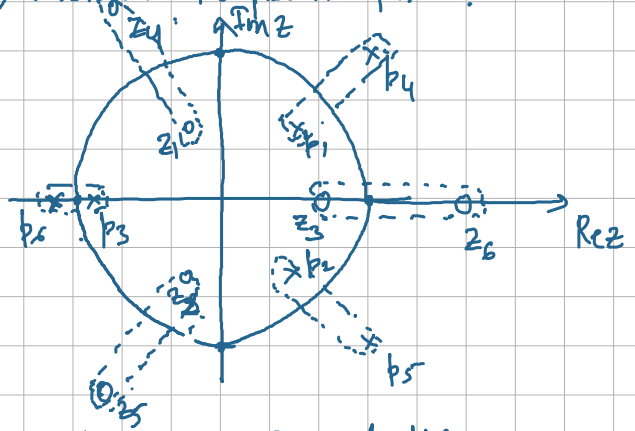
- Recap: All pass and min phase systems
- Wrap up proof
- Relating magnitude and phase response
  - | |, x, o; ∠, x, o, min-phase
  - $|H(e^{j\omega})|^2$
  - Examples

1: • $H_1(z) = \left[2(1-z^{-1})(1+0.5z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{j\pi/4}z^{-1})\right]$

• $H_2(z) = \left[(1-z^{-1})(1+2z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})\right]$

Show that $C_1(z) = C_1(z)$.

2: C(z) with poles and zeros marked below. If system is stable, causal, and a_k, b_k are real, what are the possible options?



- Recall: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M} (1-c_k z^{-1})}{\prod_{k=1}^{N} (1-d_k z^{-1})}$ = (rational system)

  Assuming b_0, a_0 are real.

  $\prod_{k=1}^{M}$ ← zeros of H(z)

  $\prod_{k=1}^{N}$ ← poles of H(z)

- $\angle H_{ap}(e^{j0}) = 0$ ← real

$$H_{ap}(z) = A \prod_{k=1}^{N_r} \frac{(z^{-1}-d_k)}{(1-d_k z^{-1})} \prod_{k=1}^{N_c} \frac{(z^{-1}-c_k^*)}{(1-c_k z^{-1})} \frac{(z^{-1}-c_k)}{(1-c_k^* z^{-1})}$$

$$H_{ap}(z)\Big|_{z=e^{j\omega}} = A \prod_{k=1}^{N_r} \frac{(1-d_k)}{(1-d_k)} \prod_{k=1}^{N_c} \frac{(1-e_k^*)(1-e_k)}{(1-e_k)(1-e_k^*)}$$

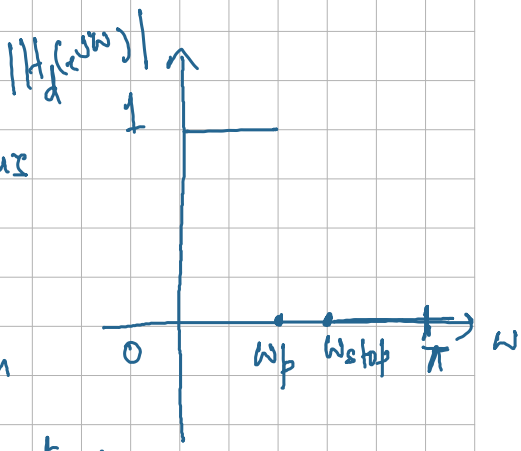
$$= A \cdot \quad \text{which is a real quantity}$$

$$\therefore \angle H_{ap}(e^{j\omega}) = 0.$$

Thus completes our proof of showing that the phase response of an all pass system (Causal, stable) is non positive.

- Recap: • Among all causal, stable, rational LTI systems $H_{min}(e^{j\omega})$ has the minimum phase-lag.
  - Similarly, $H_{min}(e^{j\omega})$ has the minimum group delay. (grpd)

- Filter design: Given $|H_d(e^{j\omega})|$, how do we design a filter to match this requirement?
- Find a function $f(\omega;\theta)$ that best solves this problem ($|\theta| = L$).
- This can be treated as a supervised learning problem and $f(\omega;\theta)$ can come from a large class of functions.



- In the following discussion we will choose a function that we have seen before: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$ i.e., the class of rational LTI systems that are stable.
- Q. Given $|H(e^{j\omega})|$ and the fact that we want to work with rational LTI systems, what can we say about $\angle H(e^{j\omega})$?
- A. In general, nothing.

- However, if causality and stability constraints are introduced, we can say more.

- $|H(e^{jw})|^2 = H(e^{jw}) \cdot H^*(e^{jw})$

$= H(z)\, H^*(1/z^*)\Big|_{z=e^{jw}}$

$= C(z)$

- Given our standard rational LTI system $H(z)$, find $C(z)$.

i.e.: $C(z) = H(z)\, H^*(1/z^*)$

$= \left(\frac{b_0}{a_0}\right)^2 \cdot \frac{\prod_{k=1} (1 - c_k z^{-1})}{\prod_{k=1} (1 - d_k z^{-1})} \cdot \frac{\prod_{k=1} (1 - c_k^* z)}{\prod_{k=1} (1 - d_k^* z)}$